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Milestone 2: Computer Vision-Based Analysis of Volleyball Serve Trajectories

Overview

Goal: Build a vision tool for volleyball serves with a fixed camera

- Detect and track the ball across frames
- Segment court lines and map boundaries
- Reconstruct serve trajectory
- Landing frame and zone
- Evaluate consistency across serves



Milestone 2 Goals

Collect and annotate 100 serves across many players

Train baseline YOLOv8 ball detector

Begin trajectory and landing frame analysis

Start CLI-interface for end user

Dataset Collection

Collection Setup:

- Single camera in the center of the court
- Approx. 5m behind the court
- Captures entire court in frame

Pipeline

- Record raw sessions -> Split into serves
- Use current model for semi-auto annotation

Current Progress

- 98 serves recorded
- 72 annotated
 - Over 10,000 frames



Model Predictions

Trained a YOLOv8s

- 736x1280 image size – may increase this for better recognition
- 50 epochs

98% precision and 95% recall

- Increase with more epochs



Milestone 2 Pitfalls

Dataset Collection

- Collecting and labeling took much longer than anticipated
- I had to delete several thousand faces
 - Early iterations of the model loved bounding faces as balls

Project Adjustments

- Simplified planned GUI to a CLI
- May remove the automatic line detection and use user input instead



Next Steps

Dataset and Model

- Update dataset with more types of balls
- Train final YOLOv8 ball detector

Evaluating trajectories

- Identify landing frame
- Use court homography to convert to real-world coordinates

Analysis

- Create trajectory plots and heatmaps

Finish user CLI



Questions

